

單因子變異數分析中的 $E(\text{MSB})$

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目標

在單因子固定效果 ANOVA 模型下證明

$$E(\text{MSB}) = \sigma^2 + \frac{\text{SSB}}{k-1},$$

其中

$$\text{MSB} = \frac{\text{SSB}^{(\text{樣本})}}{k-1}, \quad \text{SSB} = \sum_{i=1}^k n_i (\mu_i - \mu)^2.$$

注意這裡的 SSB 是以「母體平均數 μ_i 」定義的理論之間組平方和；而 $\text{SSB}^{(\text{樣本})}$ 則是以樣本平均數定義的樣本之間組平方和：

$$\text{SSB}^{(\text{樣本})} = \sum_{i=1}^k n_i (\bar{Y}_i - \bar{Y})^2.$$

我們要證明的是

$$E\left(\frac{\text{SSB}^{(\text{樣本})}}{k-1}\right) = \sigma^2 + \frac{1}{k-1} \sum_{i=1}^k n_i (\mu_i - \mu)^2.$$

模型假設與記號

考慮單因子固定效果 ANOVA 模型：

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, n_i,$$

其中

$$\varepsilon_{ij} \stackrel{\text{i.i.d.}}{\sim} (0, \sigma^2), \quad \text{且彼此獨立。}$$

記

$$N = \sum_{i=1}^k n_i, \quad \mu = \frac{1}{N} \sum_{i=1}^k n_i \mu_i \quad (\text{母體加權總平均}).$$

樣本第 i 組平均與樣本總平均分別定義為

$$\bar{Y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}, \quad \bar{Y} = \frac{1}{N} \sum_{i=1}^k \sum_{j=1}^{n_i} Y_{ij}.$$

則樣本之間組平方和為

$$\text{SSB}^{(\text{樣本})} = \sum_{i=1}^k n_i (\bar{Y}_i - \bar{Y})^2.$$

分解期望值：信號 + 雜訊

先把 $\bar{Y}_i$ 、 $\bar{Y}$ 寫成母體平均數加誤差平均：

$$\bar{Y}_i = \mu_i + \bar{\varepsilon}_i, \quad \bar{\varepsilon}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \varepsilon_{ij},$$

$$\bar{Y} = \mu + \bar{\varepsilon}, \quad \bar{\varepsilon} = \frac{1}{N} \sum_{i=1}^k \sum_{j=1}^{n_i} \varepsilon_{ij}.$$

因此

$$\bar{Y}_i - \bar{Y} = (\mu_i - \mu) + (\bar{\varepsilon}_i - \bar{\varepsilon}).$$

帶入 $\text{SSB}^{(\text{樣本})}$ 得

$$\text{SSB}^{(\text{樣本})} = \sum_{i=1}^k n_i \left[(\mu_i - \mu) + (\bar{\varepsilon}_i - \bar{\varepsilon}) \right]^2.$$

取期望值：

$$\begin{aligned} E(\text{SSB}^{(\text{樣本})}) &= \sum_{i=1}^k n_i E \left[(\mu_i - \mu) + (\bar{\varepsilon}_i - \bar{\varepsilon}) \right]^2 \\ &= \sum_{i=1}^k n_i \left[(\mu_i - \mu)^2 + 2(\mu_i - \mu)E(\bar{\varepsilon}_i - \bar{\varepsilon}) + E(\bar{\varepsilon}_i - \bar{\varepsilon})^2 \right]. \end{aligned}$$

由於 $E(\bar{\varepsilon}_i) = 0$, $E(\bar{\varepsilon}) = 0$, 故

$$E(\bar{\varepsilon}_i - \bar{\varepsilon}) = 0,$$

上述中的線性項消失，於是

$$E(\text{SSB}^{(\text{樣本})}) = \sum_{i=1}^k n_i (\mu_i - \mu)^2 + \sum_{i=1}^k n_i E(\bar{\varepsilon}_i - \bar{\varepsilon})^2.$$

再利用

$$E(\bar{\varepsilon}_i - \bar{\varepsilon})^2 = \text{Var}(\bar{\varepsilon}_i - \bar{\varepsilon}),$$

可得

$$E(\text{SSB}^{(\text{樣本})}) = \underbrace{\sum_{i=1}^k n_i (\mu_i - \mu)^2}_{\text{理論之間組平方和 SSB}} + \sum_{i=1}^k n_i \text{Var}(\bar{\varepsilon}_i - \bar{\varepsilon}).$$

因此關鍵在於計算

$$\sum_{i=1}^k n_i \text{Var}(\bar{\varepsilon}_i - \bar{\varepsilon}).$$

計算 $\text{Var}(\bar{\varepsilon}_i - \bar{\varepsilon})$

先列出各項變異與共變異：

$$\text{Var}(\bar{\varepsilon}_i) = \frac{\sigma^2}{n_i},$$

$$\bar{\varepsilon} = \frac{1}{N} \sum_{i=1}^k \sum_{j=1}^{n_i} \varepsilon_{ij} = \sum_{i=1}^k \frac{n_i}{N} \bar{\varepsilon}_i.$$

因此

$$\text{Var}(\bar{\varepsilon}) = \sum_{i=1}^k \left(\frac{n_i}{N}\right)^2 \text{Var}(\bar{\varepsilon}_i) = \sum_{i=1}^k \left(\frac{n_i}{N}\right)^2 \frac{\sigma^2}{n_i} = \frac{\sigma^2}{N^2} \sum_{i=1}^k n_i = \frac{\sigma^2}{N}.$$

另外

$$\begin{aligned} \text{Cov}(\bar{\varepsilon}_i, \bar{\varepsilon}) &= \text{Cov}\left(\bar{\varepsilon}_i, \sum_{\ell=1}^k \frac{n_{\ell}}{N} \bar{\varepsilon}_{\ell}\right) \\ &= \frac{n_i}{N} \text{Var}(\bar{\varepsilon}_i) \\ &= \frac{n_i}{N} \cdot \frac{\sigma^2}{n_i} = \frac{\sigma^2}{N}. \end{aligned}$$

因此

$$\begin{aligned} \text{Var}(\bar{\varepsilon}_i - \bar{\varepsilon}) &= \text{Var}(\bar{\varepsilon}_i) + \text{Var}(\bar{\varepsilon}) - 2 \text{Cov}(\bar{\varepsilon}_i, \bar{\varepsilon}) \\ &= \frac{\sigma^2}{n_i} + \frac{\sigma^2}{N} - 2 \frac{\sigma^2}{N} \\ &= \frac{\sigma^2}{n_i} - \frac{\sigma^2}{N}. \end{aligned}$$

將此代回上式：

$$\begin{aligned}\sum_{i=1}^k n_i \operatorname{Var}(\bar{\varepsilon}_i - \bar{\varepsilon}) &= \sum_{i=1}^k n_i \left(\frac{\sigma^2}{n_i} - \frac{\sigma^2}{N} \right) \\&= \sum_{i=1}^k \left(\sigma^2 - \frac{n_i \sigma^2}{N} \right) \\&= k\sigma^2 - \frac{\sigma^2}{N} \sum_{i=1}^k n_i \\&= k\sigma^2 - \frac{\sigma^2}{N} \cdot N \\&= (k-1)\sigma^2.\end{aligned}$$

因此

$$E(\operatorname{SSB}^{(\text{樣本})}) = \sum_{i=1}^k n_i (\mu_i - \mu)^2 + (k-1)\sigma^2.$$

令

$$\operatorname{SSB} := \sum_{i=1}^k n_i (\mu_i - \mu)^2,$$

即可寫為

$$E(\operatorname{SSB}^{(\text{樣本})}) = \operatorname{SSB} + (k-1)\sigma^2.$$

得到 $E(\operatorname{MSB})$

因為

$$\operatorname{MSB} = \frac{\operatorname{SSB}^{(\text{樣本})}}{k-1},$$

故

$$\begin{aligned}E(\operatorname{MSB}) &= E\left(\frac{\operatorname{SSB}^{(\text{樣本})}}{k-1}\right) = \frac{1}{k-1} E(\operatorname{SSB}^{(\text{樣本})}) \\&= \frac{1}{k-1} [\operatorname{SSB} + (k-1)\sigma^2] \\&= \sigma^2 + \frac{\operatorname{SSB}}{k-1}.\end{aligned}$$

得證。